BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>17</td></tr>
<tr><td>Project Title</td><td>Hospital Patient Data Analyzer</td></tr>
<tr><td>Problem Statement</td><td>Analyse hospital patient data from a CSV file to clean the data and find insights such as department-wise patient load, average length of stay and age-group distribution.
Language: Python 3 Concepts: Pandas, CSV, Data Cleaning, Grouping
Note: CSV columns: Patient_ID, Age, Gender, Department, Admit_Date, Discharge_Date. Use dummy data only.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>patients.csv:
P1,45,M,Cardiology,2026-08-01,2026-08-05
P2,30,F,Ortho,2026-08-02,2026-08-04
P3,60,M,Cardiology,2026-08-03,2026-08-08
Menu option 1 (Department count)</td><td>Cardiology: 2 | Ortho: 1
Total patients: 3</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>patients.csv:
P4,25,F,ENT,2026-08-10,2026-08-10 (same-day discharge)
Menu option 2 (Length of stay)</td><td>P4: 0 days
Average stay: 0.00 days</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>Test Case 1 rows plus:
P1 repeated (duplicate)
P6,,M,ENT,2026-08-06,2026-08-07 (age missing)</td><td>Duplicates removed: 1
Missing Age filled with median: 45</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Same CSV as Test Case 1
Menu option 3 (Stay and age groups)</td><td>Average stay: 3.67 days
Age groups: 0-40: 1 | 41-60: 2 | 61+: 0
Bar chart displayed</td></tr>
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<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Load the CSV file using Pandas.
3. Clean missing, duplicate and invalid values.
4. Use groupby() for department-wise and gender-wise analysis.
5. Calculate the length of stay and plot at least one Matplotlib chart.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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